

Sangram Kumar Sahoo

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Machine learning and perception engineer with experience building end-to-end vision pipelines for complex real-world data. Developed medical image analysis systems for angiogram stenosis detection, graph-based visual understanding models for engineering drawings, and deep learning pipelines for perception and control using PyTorch, OpenCV, and Python.

EDUCATION

Master of Science in Mechanical Engineering-Research, Carnegie Mellon University

Expected Graduation: May 2026 | GPA 3.95/4.

Subjects: AI/ML, Modern Control Theory, Soft Robots, Engineering Startup, Computer Vision, Deep Reinforcement Learning, AI Agents, Deep Learning.

Bachelor of Technology in Mechanical Engineering, SRM Institute of Science and Technology, India

Graduated: May 2023 | GPA: 8.7/10 | *Achievements: Most innovative startup ideation.*

RESEARCH EXPERIENCE

MiSUMi Corp.

Graduate Researcher, CERLAB – Prof. Kenji Shimada

(May 2025 -Present)

- Formulated engineering drawing understanding as a joint semantic and structural learning problem, predicting progressive graph cuts to hierarchically partition drawings into connected functional components.
- Integrated node semantic predictions with learned geometric embeddings to improve semantics-consistent structural decomposition in engineering shape graphs. Evaluated multiple GNN variants, including GraphSAGE and GATv2, while building custom training, debugging, and performance analysis pipelines for graph-based CAD understanding.

Komatsu Ltd.

Graduate Researcher, CERLAB – Prof. Kenji Shimada

(Aug 2024 -May 2025)

- Developed an Isaac Sim simulation environment for an SLAM test bed with the ultimate goal of creating a SLAM algorithm for a mining environment. Simulated different LiDARs and camera integration operations for the wheel loader in a simulated environment.

PROFESSIONAL EXPERIENCE

Kansai Nerolac Paints

Hosur, India

Mechanical Engineering Intern

(Jun - Jun 2022)

- Completed the daily production checklist and generated comprehensive reports on daily activities. Operated industrial machinery and acquired a thorough understanding of various manufacturing processes. Assisted in maintaining the ball mill and the daily routine of the RO water system within the manufacturing plant.

PROJECTS

Angiogram Stenosis Detection using Computer Vision

(Dec 2025)

- Developed a computer vision pipeline to analyze coronary angiogram images for vessel enhancement, artery structure extraction, and stenosis detection. Computed percentage diameter stenosis (%DS) from extracted vessel regions to support objective assessment of arterial blockage severity. Evaluated pipeline outputs through visual inspection and quantitative comparison of segmented vessels and detected narrowing regions.

Reinforcement Learning-based neural architecture search for deepfake voice detection

(May 2025)

- Established a comprehensive data processing pipeline for deep learning based audio preprocessing- feature extraction
- Implemented and evaluated baseline models, achieving an Equal Error Rate (EER) of 2.8% with 50 epochs
- Developed an initial proof-of-concept for the Neural Architecture Search framework (currently under refinement)

Generative Trajectory Aggregation for Imitation Learning in CarRacing-v2

(Oct 2025)

- Co-developed a diffusion-based DAgger alternative that replaced online expert queries with a generative teacher trained on offline PPO demonstrations.
- Built an end-to-end pipeline with trajectory diffusion modeling, synthetic corrective action generation, and CNN student policy training for robust visuomotor control.
- Achieved 794.4 mean episodic return across 20 evaluation episodes, outperforming standard behavior cloning in long-horizon driving tasks

SKILLS

Frameworks: Python (PyTorch, TensorFlow, CUDA, PySpark, OpenCV), C++, Docker, ROS, Kotlin.

Database & Cloud: Google Cloud Platform (GCP), AWS, Power BI.

Software: Solidworks, Autocad, Matlab

Leadership: Youth India Foundation, Co-Founder.